Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An LED’s forward voltage is…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always zero

B. Equal to any source

C. Device- and condition-dependent

D. Always 0.7 V

Correct: C

2. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Correct: C

3. (2 points) Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. No voltage exists

B. Opposite the assumed direction

C. Impossible circuit necessarily

D. Negative charge disappears

Correct: B

4. (2 points) A diode is generally…

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. A mechanical switch

B. Nonlinear

C. An ideal fixed resistor

D. Always conducting both ways

Correct: B

5. (4 points) Why is charge not consumed?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Steady charge flow conserves charge while energy changes form. (4 points).

6. (4 points) Write a source-two-resistor loop equation.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: Vs - IR_1 - IR_2 = 0 for traversal with current. (4 points).

7. (4 points) Name two LED faults.

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Reversed orientation and an open path; node measurements can distinguish them. (4 points).

8. (4 points) Why is Ah not sufficient for battery energy?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Voltage and operating conditions are also required. (4 points).

9. (4 points) Explain a nodal branch current.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: Current from node a to b through R is (Va - Vb)/R. (4 points).

10. (4 points) Why not infer brightness from current alone?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

answer: Efficiency, optics, temperature, and device characteristics affect light output. (4 points).

11. (2 points) A 9 V ideal battery drives a 4.5 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 5 and 7 ohms. Find equivalent resistance.

answer: 12 ohms (2 points).

Numerical target 12; absolute tolerance 0.024. Accepted units: , ohms.

13. (2 points) Two identical 10 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 5 ohms (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 9 V. Find power.

answer: 18 W (2 points).

Numerical target 18; absolute tolerance 0.036. Accepted units: , W.

15. (2 points) A node has incoming currents 6 A and 2 A and one outgoing current. Find the outgoing current.

answer: 8 A (2 points).

Numerical target 8; absolute tolerance 0.016. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

answer: In the ideal model the reverse-biased diode blocks current; a forward-biased conducting diode permits conventional current from anode toward cathode. Real leakage and forward voltage require a stated device model. (2 points).

19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).